ANNALISE DANTES

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EDUCATION

University of Florida - Herbert Wertheim College of Engineering **May 2029**
Bachelor of Science in Mechanical Engineering

Relevant Coursework: Multivariable Calculus, Computer Aided Graphics and Design, Computational Linear Algebra, Python Programming for Engineers, Engineering Mechanics: Statics, Design and Manufacturing Laboratory

PROFESSIONAL EXPERIENCE

Edison Township Board of Education - *Technology Intern* **June 2025 – June 2026**

- Diagnosed, repaired, and configured MacBooks and classroom technology for district-wide deployment.
- Performed hardware and software troubleshooting, operating system setup, and device imaging.
- Collaborated with IT personnel to support technology rollouts and resolve technical issues efficiently.

Talaria Research Institute - *Astrophysics Research Scholar* **May 2025 – Sept 2025**

- Authored a research paper on Coronal Mass Ejection Kinematics alongside Professor Diana Besliu-Ionescu of the Romanian Astronomical Institute and Marilena Mierla of the Royal Observatory of Belgium.
- Analyzed images from the SOHO/LASCO instrument using JHelioviewer to observe coronal mass ejection (CME) propagation and data from the CDAW CME catalog.
- Calculated CME kinematics and compared results with data on high-frequency radio disruption on Earth to determine the extent of CME-induced geomagnetic storms' interference with radio waves.

New Jersey City University and Liberty STEM Ecosystem - *Teaching Intern* **July 2024 – Aug 2024**

- Supported hands-on STEM instruction spanning laboratory techniques, mathematics, and engineering design.
- Trained under a supervising professor in laboratory procedures before transitioning to independent activity leadership.
- Guided students through microscope-based cell observation, logic and problem-solving activities, and engineering projects.

LEADERSHIP/ACTIVITIES

Swamp Launch (Solid Rocketry) - *Payload Mechanics Subteam and Structures Subteam Member* **Aug 2026 – Present**

- Designing and building the physical payload structure, including component mounting, a sensor suite, and parachute deployment systems.
- Leading the new member project structures subteam in designing and building an L1-class rocket through component selection, structural design, and testing.
- Receiving machining training to support manufacturing in the spring semester

Gator Motorsports (Formula SAE) - *Aerodynamics Subteam Member* **Aug 2026 – Present**

- Contribute to vehicle aerodynamic design through CAD modeling, analysis, and iterative development of aerodynamic components.
- Currently designing a front wing using SolidWorks and STAR-CCM+ CFD simulations to practice evaluating aerodynamic performance and analyzing downforce, drag, and airflow characteristics.

PROJECTS

Custom Quadcopter Build | In Progress **2026-Present**

- Integrating propulsion, power, and flight-control components into a custom quadcopter platform.
- Researching hardware for part selection and configuring INAV flight-control software.
- Exploring GPS-assisted navigation and autonomous flight capabilities for future development.

River Pollutant Modular Treatment Systems (Plantchar) **2024-2025**

- Utilized a $7.5k grant through the MIT Lemelson InvenTeams Initiative to address pollution in the local Raritan River.
- Designed and tested two 3-D printed prototypes: a biochar (pollutant absorbent) filled filtration pod and a floating wetland treatment using plant root pollutant uptake
- Assessed strengths and weaknesses of prototypes and synthesized them into a final prototype; a hybrid modular treatment system combining both methods to maximize pollutant absorption, meeting NJDEP water quality standards during testing

SKILLS/CERTIFICATIONS/AWARDS

Technical Skills: STAR-CCM+, SolidWorks, Fusion 360, Python, JavaScript, MATLAB, ANSYS, Microsoft Office (Excel, Word, Powerpoint), Arduino, 3D Printing, HTML
Awards: National Merit Award Recipient, Naval STEM Horizons Highest Honors, ICDC Finalist, University Merit Scholar